

Devastating outbreak of bark beetles in the Czech Republic: Drivers, impacts, and management implications

T. Hlásný^{a,*}, S. Zimová^a, K. Merganičová^a, P. Štěpánek^b, R. Modlinger^a, M. Turčáni^a

^a Czech University of Life Sciences in Prague, Faculty of Forestry and Wood Sciences, Czech Republic

^b Global Change Research Institute, Czech Academy of Sciences, Czech Republic

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ABSTRACT

Outbreaks of tree-killing insects are intensifying globally, affecting economies, human well-being, and driving ecosystem transitions. The Czech Republic has recently become Europe's epicenter of the outbreak of spruce bark beetle *Ips typographus*, the most aggressive species in Eurasia. We investigated a countrywide outbreak dynamic during the period 2003–2019, with a special focus on the period 2017–2019 when the outbreak reached an unprecedented intensity. In order to identify main outbreak drivers, we investigated annual time series of the volume of trees killed by bark beetles in the Czech districts ($n = 77$), and a suite of climatic and forest structure-related predictors using Generalized Additive Models. Finally, we reviewed a large body of public materials to understand broader social, ecological, and economic implications of the outbreak.

We found that bark beetles were damaging 0.2–1.4% of Norway spruce growing stock annually across the Czech Republic in the period 2003–2016. This level increased to 3.1–5.4% in 2017–2019, causing the total depletion of spruce in some regions. The long-term bark beetle dynamics (2003–2019) was driven by the combination of wind disturbance and climatic drivers, represented in our study by annual temperature anomaly and Standardized Precipitation-Evapotranspiration Index. However, the effect of wind was diminished during the period 2017–2019, whereas the effect of drought dominated. Our findings thus suggest a transition from wind- to drought-driven bark beetle dynamics.

The outbreak and subsequent large-scale salvaging and wood transportation affected quality of life of people in a broad vicinity of outbreak areas. Extensive management actions aggravated some of the notorious conflicts between forest management and nature conservation, and highlighted the poor harmonization of respective policies. A decrease in timber price, an excessive workload, and other cascading effects caused severe revenue loss, requiring state interventions amounting to ca 260 million EUR in 2018–2019.

We suggest that increasing frequency of climate extremes in combination with the unfavorable forest structure pushed Central European spruce forests to the margin of their ecological space and unfolded large-scale forest transformations. Effective responses will require fundamental structural changes in the regional forest-based sector, particularly aiming at increased social and ecological resilience.

1. Introduction

Tree killing bark beetles are prominent disturbance agents in temperate and boreal biomes and affect the structure, functioning and composition of the forests (Vindstad et al., 2019). In Europe, spruce bark beetle *Ips typographus* (L.) is an economically significant forest insect pest, attacking primarily mature Norway spruce *Picea abies* (L.) H. Karst trees (Christiansen and Bakke, 1988). As Norway spruce is a cornerstone species of the timber industry in many countries, such outbreaks may have severe impacts on forestry economies, with cascading effects

affecting the entire forest-based sector and international timber markets (Grégoire et al., 2015; Montagné-Huck and Brunette, 2018). The outbreaks also compromise numerous regulatory and cultural ecosystem services, including forest carbon storage as well as aesthetic and recreational values (Dobor et al., 2018; Thom and Seidl, 2016). The recent outbreaks in Europe have therefore generated serious concerns about the stability of timber markets, effects on the environment, and human wellbeing (Hlásný et al., 2019; Morris et al., 2017).

Outbreaks of *I. typographus* have intensified in recent decades and significantly contributed to the observed doubling in canopy mortality

* Corresponding author.

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in Central Europe (Seidl et al., 2014; Senf et al., 2018). Climate is an important driver of such dynamics due to the ectothermic physiology of bark beetles (Jakoby et al., 2019). Climate controls the timing of bark beetle phenology, number of annual generations, and winter mortality (Baier et al., 2007; Marini et al., 2017). Climate also affects the conditions of host trees and their susceptibility to bark beetle attacks (Huang et al., 2020; Netherer et al., 2015). Interactions between the climatically sensitive development of bark beetles, the fitness of host trees, and the availability of breeding substrate determine whether bark beetle populations can cross the epidemic threshold and unfold the outbreak (Raffa et al., 2008). Recent changes in climate, which included an increased air temperature, changed precipitation patterns, and an increased frequency of heat and drought spells have thus critically amplified bark beetle outbreaks in Europe (Marini et al., 2017). Climate extremes not only intensify the outbreaks but also cause their large-scale synchrony (Seidl et al., 2016a). Model projections indicate that outbreak intensity will continue to increase in the coming decades, causing a large-scale transformation of forest landscapes with implications for the entire forest-based sector (Albrich et al., 2020; Dobor et al., 2020a; Honkaniemi et al., 2020). In Europe climate change is expected to increase the level of bark beetle disturbance sevenfold up to 2030 compared to the period 1971–1980 (Seidl et al., 2014). Other studies have suggested an increase in bark beetle disturbance during the 21st century by 60–220%, depending on the level of climate forcing and forest conditions (e.g. Dobor et al., 2020a, 2020b; Sommerfeld et al., 2020).

The other factor driving the recent outbreak intensification is the historical transformation of the forest structure (Munteanu et al., 2015; Seidl et al., 2011; Sommerfeld et al., 2018). In Central Europe, Norway spruce has been extensively planted outside the range of its native distribution at the expense of other, mostly broadleaved species to meet the requirements of expanding industries (Klimo et al., 2000; Spiecker et al., 2004). The prevalence of clear-cutting and long-term prevention of forest disturbances further homogenized the forest structure and increased the level of forest growing stock (Dobor et al., 2020a). Such management created large forest areas susceptible to an array of disturbances with eroded resilience mechanisms (Hanewinkel et al., 2008). The coincidence of such an unfavourable forest structure with erratic changes in climate may cause the limits of ecological resilience (tipping points; Dakos et al., 2019) to exceed and large-scale forest transformations to unfold (Albrich et al., 2020; Scheffer et al., 2009; Turner et al., 2020). Sommerfeld et al. (2018) suggested that climate change and historical forest management contributed equally to the recent outbreak intensification in Europe.

Although bark beetle outbreaks have been intensifying in Europe for more than two decades (Seidl et al., 2014), the size of the very recent spruce mortality pulse was unprecedented (Senf et al., 2020; Schuldt et al., 2020). This outbreak eruption was observed in several countries, including Austria, Germany, and France (e.g. Jandl, 2020; Senf and Seidl, 2020; Sommerfeld et al., 2020), with the Czech Republic becoming an epicenter (Hlásny et al., 2019; Toth et al., 2020). The most recent outbreak in the Czech Republic started in ca. 2015 and exceeded ranges observed over the last two centuries. The outbreak received various connotations such as the ecological collapse or failure of spruce-oriented management. Large pulses of salvaged timber had overarching effects on the regional forest-based sector and raised a great interest among the public, ecologists, timber industry experts and policy-makers in understanding outbreak mechanisms and prospects. Understanding recent changes in disturbance regimes is also important for developing management strategies that are able to address the emergent ecosystem dynamics, such as resilience-oriented stewardship (Senf and Seidl, 2020).

In response to these challenges, we aimed to investigate here the spatial and temporal dynamics of bark beetles in the Czech Republic during the last 20 years, focusing particularly on the period 2017–2019, which was characterized by an unprecedented outbreak intensity. With regard to the cascade of impacts triggered by the outbreak, we further

aimed to understand major implications on forest owners, the environment, and the society and how different actors responded to the outbreak and its management. We hypothesized that the outbreak was triggered by the extremely dry climate in recent years, which replaced the role of wind as the notorious outbreak trigger in Central Europe. We further hypothesize that the outbreak triggered a cascade of processes involving different sectors and social groups, resulting in tensions that constrained management efforts. Such an investigation can help one to better understand the broader social context of the outbreak and inform future management decisions.

2. Data and methods

2.1. Forests and forestry in the Czech Republic

The Czech Republic belongs to the temperate forest zone of Europe with a temperate oceanic climate (Rivas-Martínez et al., 2004). Forests cover 2.67 million ha, representing 34.1% of the land area. Intensive management applied over the last three centuries resulted in the dominance of Norway spruce that represents 50% of the current tree species composition. Scots pine *Pinus sylvestris* (16.2%) and European larch *Larix decidua* (3.8%) are other abundant conifers. Broadleaves constitute 27.3% of the species composition, with dominance of the European beech *Fagus sylvatica* and oaks *Quercus* sp. Both dominating conifers, spruce and pine, have been suffering from mass mortality due to drought and biotic attacks in recent years. Forests under the state ownership represent 55% of the total forest area, while the non-state forests (private, church, and municipal) are owned by ca 300,000 different legal entities. Small-scale clear-cutting (1 ha area) with a mean forest rotation age of 115 prevails (Ministry of Agriculture of the Czech Republic, 2019).

2.2. Studied disturbance system

The investigated outbreak was caused by the European spruce bark beetle *Ips typographus* (L), which is the main biotic disturbance agent in Norway spruce forests in Eurasia (Biedermann et al., 2019). *I. typographus* attacks spruce trees to access fresh phloem needed for brood development. Successful colonization is typically fatal to trees, because beetles destroy the inner bark and thus disrupt nutrient transport to roots. *I. typographus* causes intermittent outbreaks with devastating effect on forest landscapes, which are typically interrupted by long non-outbreak periods (Raffa et al., 2008). In the non-outbreak period when beetle populations are small, beetles avoid tree defense by entering only freshly dead or stressed trees (Huang et al., 2019). During the outbreaks, however, bark beetles can exhaust tree defense by mass coordinated attacks. While factors triggering the outbreak are relatively well understood, factors leading to outbreak collapse are subject of scientific debate (e.g. Biedermann et al., 2019).

In Central Europe, population transition to the outbreak phase is typically triggered by windthrows, which provide large amounts of breeding substrate (Mezei et al., 2017; Stadelmann et al., 2014). Notorious outbreak areas where wind and bark beetles represent a typical disturbance regime are, for example, the Bavarian Forest in Germany (van der Knaap et al., 2020), the High Tatras in Slovakia (Mezei et al., 2017), and the Bohemian Forest in Czechia (Zemek et al., 2003). Outbreaks can also be triggered by other factors compromising tree defense, particularly hot and dry weather (Marini et al., 2017; Netherer et al., 2019).

I. typographus exhibits large phenological plasticity in thermally-regulated traits and this allows it to adjust the number of annual generations and generation timing to local climates (Bentz et al., 2019). A warming climate therefore not only compromises tree vitality and defense, but it also reduces the winter mortality of beetles and allows the completion of additional beetle generations per year (Baier et al., 2007; Berec et al., 2013). Recent observations from the Czech Republic

indicate a shift to a three-generation regime in some locations, likely contributing to the recent increase in tree mortality.

Bark beetle management follows two major objectives, i.e. outbreak prevention and outbreak containment (Wermelinger, 2004). While prevention mainly aims at keeping beetle densities low via sanitation removal of infested trees, beetle trapping, application of pesticides, and the swift removal of wind-felled trees, containment mainly aims at massive salvage and sanitation logging in outbreak areas or their surroundings to prevent outbreak expansion (Faccoli and Stergulc, 2008; Holuša et al., 2017). The efficiency of these strategies, however, has recently been increasingly questioned (Dobor et al., 2020b, 2019; Hlásny et al., 2019).

2.3. Data

To characterize spatial and temporal dynamics of bark beetle disturbance, we combined several sources of data on forest damages reported by forest owners. First, we used annual reports of the Czech Forest Protection Service (FPS) from the period 2003–2019. The reports include amounts of salvaged trees (m^3) killed by different agents. Strict legal obligation to salvage dead trees suggests a high correspondence of killed and salvaged trees, yet the salvaging rate may change depending on local circumstances. We extracted “bark borers on spruce” category from the FPS statistics. Vast majority of killed trees in this category accounts for *I. typographus*, though other species such as *I. duplicatus*, *I. amitinus* and *Pityogenes chalcographus* can be locally important (Holuša et al., 2012, 2010). The data are reported for administrative districts at the LAU1 level of the EU nomenclature (Local Administrative Units, $n = 77$). These reports are mainly based on data reported by the Forest of the Czech Republic and the Military Forests and Properties, and cover ca 70% of the total forest area of the country. The completeness of the data (i.e. part of the total forest area covered by the source reports from forest owners), however, varies between years.

Second, we used annual reports of the Czech Statistical Office (CSO), which contain forest damage records reported by all economically active entities in the Czech Republic at the level of NUTS3 (Nomenclature of Units for Territorial Statistics, EUROSTAT, $n = 14$). Therefore, while spatial resolution of these data is lower than that of the FPS data, they cover a larger forest area. Consistently with the FPS data, we extracted annual data on bark-beetle-related salvage logging for the period 2003–2019 from the CSO statistics.

Both data sources (FPS and CSO) are derived by aggregating primary data reported by forest owners. The source units of the reporting are rather heterogeneous, depending on the ownership structure, and are not consistent with the boundaries of administrative districts. The primary source data, however, are not publicly available.

Next, we combined the two data sources to create a dataset with coverage of CSO, and the resolution of FPS. First, we summed salvaged volumes in LAU1 districts (FPS) pertaining to each NUTS3 district (CSO), yielding 14 pairs of salvaged volumes for each year ($\Sigma LAU1_{ij}$ vs. $NUTS3_j$, where $i = 1-77$ (LAU1) and $j = 1-14$ (NUTS3)). The ratios of these value pairs yielded a set of coefficients, which were applied on the LAU1 salvaged volumes to ensure that they reached the levels of the CSO statistics. Such a procedure preserved the spatial pattern of salvaged volumes reported by the FPS (LAU1 level), while matching the sums of salvaged volumes in the CSO data (NUTS3 level).

To reach a more realistic allocation of differences between LAU1 districts pertaining to the specific NUTS3 district, we calculated the proportions of spruce in species composition in each LAU1 relative to the spruce growing stock in the respective NUTS3 (Source: Forest Management Institute of the Czech Republic), and redistributed the respective differences accordingly. Hence, the total salvaged volumes from the CSO statistics were preserved, and the distribution of values across the LAU1 districts was refined by considering the actual distribution of Norway spruce. The final dataset that was subject to further analyses consisted of 77 time series (2003–2019) associated with the 77

LAU1 districts.

The third source of data used was data on actual spruce growing stock in all years of the study period. These data were extracted from the Forest Management Plan (FMP) database supervised by the Forest Management Institute of the Czech Republic. The data are being updated in the field in a 10-year cycle, so they are not temporally consistent with the previously described annual damage data. We therefore modified the growing stock by salvaged volumes on an annual basis, with regard to the year when the FMP data for the largest forest area within an LAU1 district were updated. This modification was particularly important in districts where the outbreak has recently culminated, and spruce growing stock was reduced within a short period of time. We also adjusted the actual growing stock levels by the estimated volume increment and regular harvests. Here we note that regular harvests have been substantially reduced during the recent outbreak period (2018–2019) to mitigate impacts on the timber market (Toth et al., 2020).

Climate data were derived from weather stations supervised by the Czech Hydrometeorological Institute. Data quality control and homogenization were conducted based on the methods described by Štěpánek et al. (2013). Source daily values were aggregated into monthly, seasonal and annual means (air temperature) or sums (precipitation). The monthly aggregates were interpolated by means of the regression kriging into the maps with a spatial resolution of 500 m. Various terrain characteristics were used as predictors in the interpolation. The monthly maps were then used to calculate different annual statistics to be used as predictors in the developed regression models (Appendix A) for all districts ($n = 77$). The Standardized Precipitation-Evapotranspiration Index (SPEI) was calculated based on the method described by Vicente-Serrano et al. (2010) and Beguería et al. (2014). The calculation was conducted using the R package SPEI (Vicente-Serrano et al., 2010) and ProClimDB software (Štěpánek, 2010).

2.4. Regression modelling design and data

To identify the drivers of bark beetle dynamics over the period 2003–2019, we formulated a statistical model based on the Generalized Additive Models (GAMs) (Wood, 2017). GAMs are used for modelling complex regression functions and gained an increased recognition in ecological research (Pedersen et al., 2019). GAMs approximate the relationship between the response variable and the predictors by means of several non-linear smooth functions, which are subject to inferences and interpretations. GAM fitting aims to determine the optimal smoothing parameters, while controlling for the model's overfitting. GAMs are formally defined as:

$$g(E(Y)) = \alpha + s_1(x_i) + \dots + s_p(x_p)$$

where Y is the response variable, $E(Y)$ is the expected value, and $g(Y)$ is the link function that links the $E(Y)$ to the predictor variables $x_i, \dots, x_p$. The terms $s_i, \dots, s_p$ are nonparametric smooth functions.

The first response variable used was the percent of spruce growing stock annually affected by bark beetles and salvaged (Kill%) in each year of the period 2004–2019. The number of records used to estimate the model parameters was given as 77 LAU1 districts $\times$ 16 years (2004–2019; the year 2003 was omitted to allow for including predictor variables shifted by one year backward, such as the wind-felled volumes from the previous year). Second, we investigated Kill% over the recent period 2017–2019, when the level of disturbance culminated (Fig. 1) (i.e. three years $\times$ 77 districts). Third, we investigated the slope of the linear regression of Kill% over the period 2017–2019 (Slope; one value $\times$ 77 districts) (Appendix A). In the latter case, we considered forest and climate conditions from the year 2018 as predictors.

The candidate predictor variables included the long-term annual mean temperature and precipitation total characterizing the position of districts along the major climatic gradients in the country, the mean

forest age and proportion of spruce in the districts, the volume of salvaged wind-felled trees, the annual temperature and precipitation anomalies (difference from the long-term average 1980–2010), and the annual Standardized Precipitation-Evapotranspiration Index (SPEI; Thornthwaite, 1948) (Appendix A). The volume of salvaged windfelled trees needs to be interpreted as the relative intensity of wind disturbance rather than the availability of breeding substrate, as it represents the amount of removed rather than retained windfelled trees. In case of windfelled volumes and annual climatic predictors, we considered values from both the actual and the preceding year. We used a step-wise selection to identify the predictors to be included in the model.

The Kill% was modelled using a Tweedie distribution (Wood et al., 2016) with the log link function. The Slope was modelled using a Gaussian distribution with the log-link function. A thin plate regression spline was fit to each predictor variable. Selected interactions between the predictors based on the tensor product approach were included in the models, too.

We constructed several statistical models and tested their performance using the Akaike Information Criterion, by evaluating the statistical properties of residuals, and the amount deviance explained (analogous to variance in a linear regression) by the models. We used R library mgcv (Wood, 2017) to conduct the presented analyses.

2.5. The Pettitt test

We used the Pettitt test (Pettitt, 1979) to identify years where Kill% started to increase in the LAU3 districts. The Pettitt test is a non-parametric test that identifies change points (years of significant change in time series of Kill% in this study) in a univariate time series based on the Mann-Whitney statistic. We used the R package trend v.1.1.2. for this analysis (Pohlert, 2020).

3. Results

3.1. Spatial and temporal patterns

Total volume of trees killed by bark beetles and salvaged across the Czech Republic oscillated around 1.5 million m³ annually in the period 2003–2015, representing 0.2% of the total growing stock (Fig. 1a). Years 2015 and 2016 exhibited substantial increases in the disturbance rate, reaching 2 and 4 million m³ annually, respectively. Most dramatic

increase occurred in 2018, reaching 13 million m³. In 2019, the amount of salvaged trees killed by bark beetles increased further, reaching nearly 23 million m³ (3.2% of the total growing stock). Spruce growing stock decreased from 511 ± 15 mill. m³ in 2011–2014 to 430 ± 15 million m³ in 2019 (–16%) (Adolt et al., 2020).

While trees killed by bark beetles represented 10–49% of the total salvaged volume in the period before 2018, they dominated in the period 2018–2019, accounting for 56–73% of the salvaged stock (Fig. 1a). At the same time, while salvaged volume due to bark beetle attacks represented 5–30% of the total harvests before 2018 (i.e. salvaged and planned), it accounted for 50–70% in 2018–2019. Moreover, total salvaged volume (caused by all disturbance agents) in 2018–2019 represented 90–95% of the total harvests; i.e. planned harvests were greatly reduced for logistic and market reasons.

Kill% started to increase in most of the districts in 2014, while the increase was delayed in several districts in the western part of the country by 2–3 years (Fig. 1bd). Decreasing Kill% in the period 2017–2019 was observed only in four districts in the north-east (Fig. 2), where the outbreak that has persisted there since the 1990s (Hlásny and Šitková, 2010) already depleted spruce stocks, resulting in the opposite trend to the rest of the country.

Bark beetle disturbance was scattered across the country before 2017, remaining below 0.5% of the growing stock affected (and salvaged) annually (Fig. 2). The outbreak expanded dramatically in 2018–2019, culminating in the southeastern districts. The highest acceleration of the outbreak indicated by the slope of linear regression in Kill% over the years 2017–2019 was observed in the south-eastern districts too.

The outbreak has not yet affected the high-elevation areas in the northwest of the country with a high spruce proportion (grey areas in Fig. 2). These districts were severely affected by air pollution in the 1970s and 1980s, shifting forest age structure towards younger and less vulnerable stages. Moreover, these areas are characterized by a colder and wetter climate than most of the remaining districts. In particular, long-term (1980–2010) average annual precipitation total in districts depicted in grey in Fig. 2 was 775 mm, while it was 670 mm in the rest of the country. The same comparison for temperature is 7.3 °C vs. 8.1 °C.

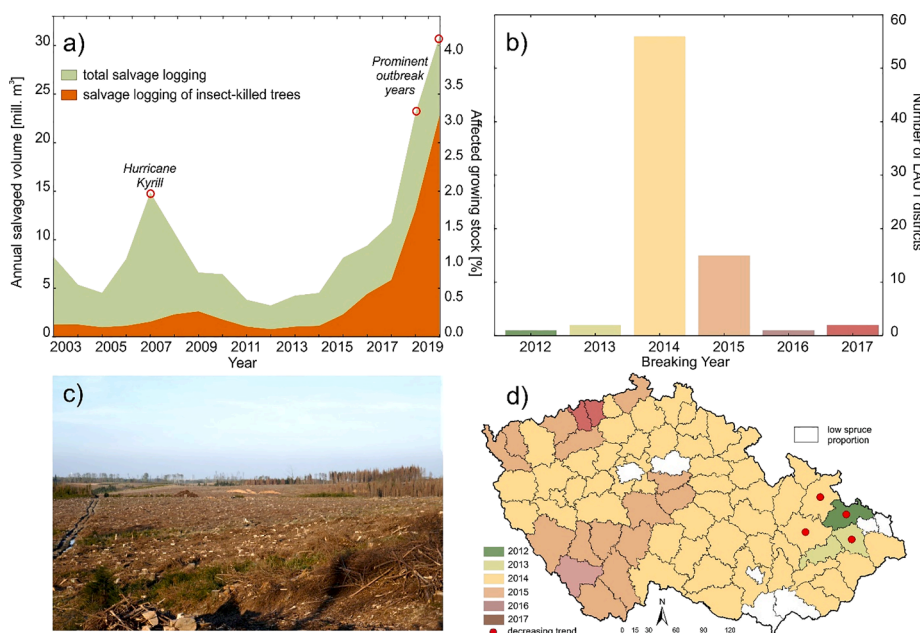


Fig. 1. a) Total annual salvaged volume and salvaged volume of trees killed by bark beetles in the Czech Republic over the period 2003–2019 (left axis). The right axis shows an approximate percentage of affected growing stock. Major disturbance events are indicated too. b) Distribution of years with a significant change in the times series of Kill% in the districts of the Czech Republic as indicated by the Pettitt test. c) A post-outbreak salvaged area (North-eastern Czechia, the Olomouc county). d) Spatial distribution of the years of significant change in time series indicated in the panel b). Red dots indicate districts where the disturbance rate decreased after the year of significant change. White areas indicate districts with low spruce proportion, which were excluded from the investigation. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

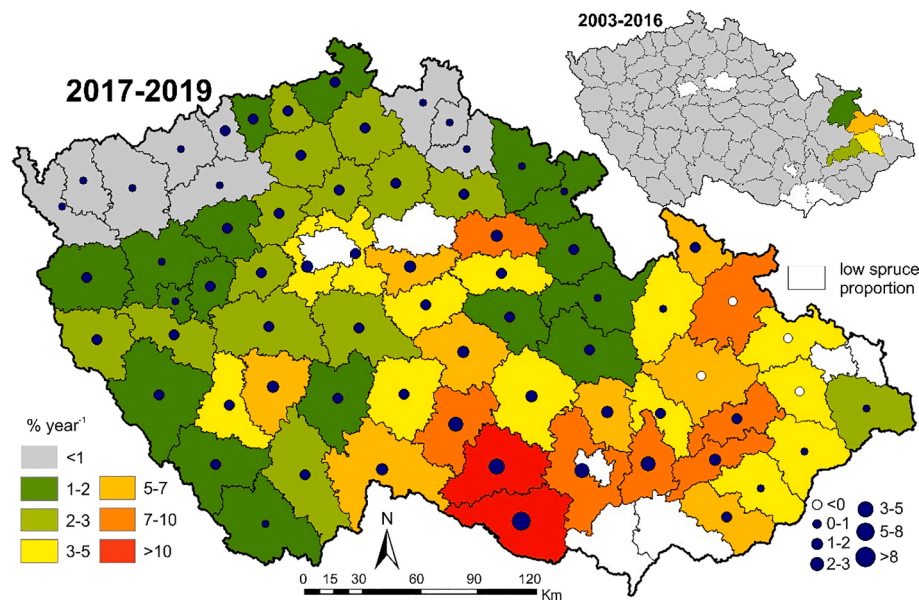


Fig. 2. Average percent of spruce growing stock affected by bark beetles and salvaged over the period 2017–2019. The circular symbols show the slope of the linear regression over this period. The insert map shows the situation in the preceding period 2003–2016.

3.2. Drivers of bark beetle dynamics

3.2.1. Longer-term perspective

We retained in the analysis of bark beetle dynamics in the period 2004–2019 the following predictor variables: the mean annual air temperature in the period 1980–2010 (T), the proportion of spruce (Spruce%), the mean spruce age (Age), the amount of windfelled and salvaged wood in the preceding year ($Wind_1$), the temperature anomaly ($Tanom$), and the Standardized Precipitation Evapotranspiration Index

(SPEI) (Appendix A). The data represent average values for the LAU1 districts ($n = 77$). Moreover, we included the interaction term Spruce% $\times$ Age. The model explained 73% of deviance of Kill%. All of the predictors except for Age were found to have a smoothing term significantly different from zero ($p < 0.001$). We, however, also retained the Age variable as its interaction with Spruce% was significant, and markedly increased the amount of explained deviance (Fig. 3).

Response of Kill% to T had a U-shaped pattern, indicating that bark beetles were affecting forests across a broad range of temperature

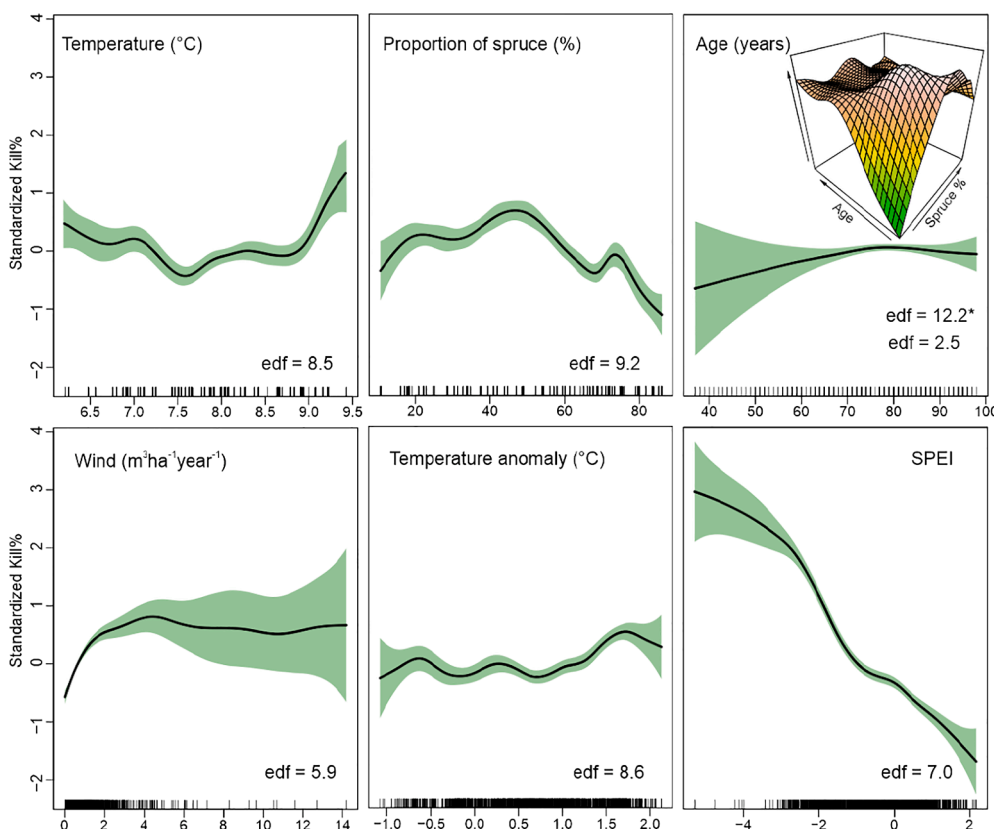


Fig. 3. Response of Kill% in the 77 districts of the Czech Republic in the period 2004–2019 to selected predictors modelled by means of the Generalized Additive Models. A 95% confidence interval is indicated for each smooth term. The insert in the Age panel indicates the interaction of Spruce % and Age. Marks along the horizontal axis represent single observations. edf – effective degrees of freedom. *indicates edf for the interaction term. SPEI – Standardized Precipitation-Evapotranspiration Index.

conditions. Still, there is an increased level of disturbance in districts with the mean annual air temperature above 9 °C. A unimodal response to Spruce% culminating at 50% proportion of spruce in the districts reflects the fact that the districts with high spruce proportions located mostly in the western part of the country have not yet been affected by bark beetles. The response of Kill% to Age was indistinct, though slightly increasing with an increasing age. Kill% increased in response to wind disturbance intensity (approximated by the volume of salvaged wind-felled trees) from the previous year ($Wind_{-1}$). This increase was distinct at low levels of wind impact, whereas it remained stable under higher windthrow intensities. Response of Kill% to Tanom was indistinct up to the value of +1.5 °C, and then increased. The response of Kill% to SPEI was found to be most distinct. Low SPEI values indicating reduced water availability correspond with the highest levels of Kill% and vice versa. The lowest values of SPEI and the highest values of Tanom correspond with the period of recent outbreak intensification from 2018 to 2019 (Appendix B).

3.2.2. Perspective of the recent outbreak culmination

Further, we investigated the drivers of Kill% in the outbreak period 2018–2019, and the drivers of outbreak acceleration over the period 2017–2019 as indicated by the slope of the linear regression of Kill% (Slope) (Fig. 4). The models explained 64% and 66% of deviance in Kill% and Slope, respectively. Both Kill% and Slope were positively associated with warmer (T) and drier conditions (SPEI). Kill% was affected by forest age, while the outbreak accelerated more intensively in districts with a higher spruce proportion. Interestingly, Age did not affect Slope and Spruce% did not affect Kill%. Finally, higher levels of Kill% were associated with districts experiencing an excess of precipitation (Panom), a relationship lacking an ecological interpretation. Wind-related variables did not have any effect on either response variable in this recent period.

3.3. Outbreak impacts and responses

The outbreak and extensive counter-measures triggered a cascade of processes affecting different sectors and social groups. We evaluated a large number of media articles, annual reports of state administration, forest enterprises and NGOs to identify four broad categories of impacts, which required different responses from the state administration and forest management (Table 1).

A number of sources have indicated negative impacts on the quality of life and the safety of inhabitants in the affected areas. These impacts included a risk of injuries from falling trees and forest mechanisms as well as the heavy traffic of trucks with salvaged timber, damage to the roads and an increased risk of traffic accidents. The outbreak reduced forest recreation and scenic values over an area of up to ca. 55,000 ha, including ca. 13,000 ha of protected areas (national parks, protected landscapes, etc.). Measures taken to mitigate these impacts included communication campaigns, an extensive clearing and reforestation of disturbed areas, investments into roads reconstruction, and the adoption of new regulations harmonizing the need of excessive transportation with impacts on the quality of life of affected inhabitants (see Table 1).

Economic implications of the outbreak were mainly related to the excessive workload related to the salvaging and reforestation of cleared areas as well as the decrease in timber price (Table 1, Appendix D), causing an economic decline of many forest owners and enterprises. Alleviating these impacts required state interventions in terms of subsidy, compensation and recovery payments amounting for ca. 260 million EUR in the period 2018–2019. The estimated amount of these payments for 2021 is 267 million EUR (Ministry of Agriculture of the Czech Republic, 2020, 2019).

The effect of management on areas under different level of protection was identified as another important outbreak implication. The management included a heavy salvaging of dead trees and the use of insecticides in storage yards, and it often did not respect processes such as

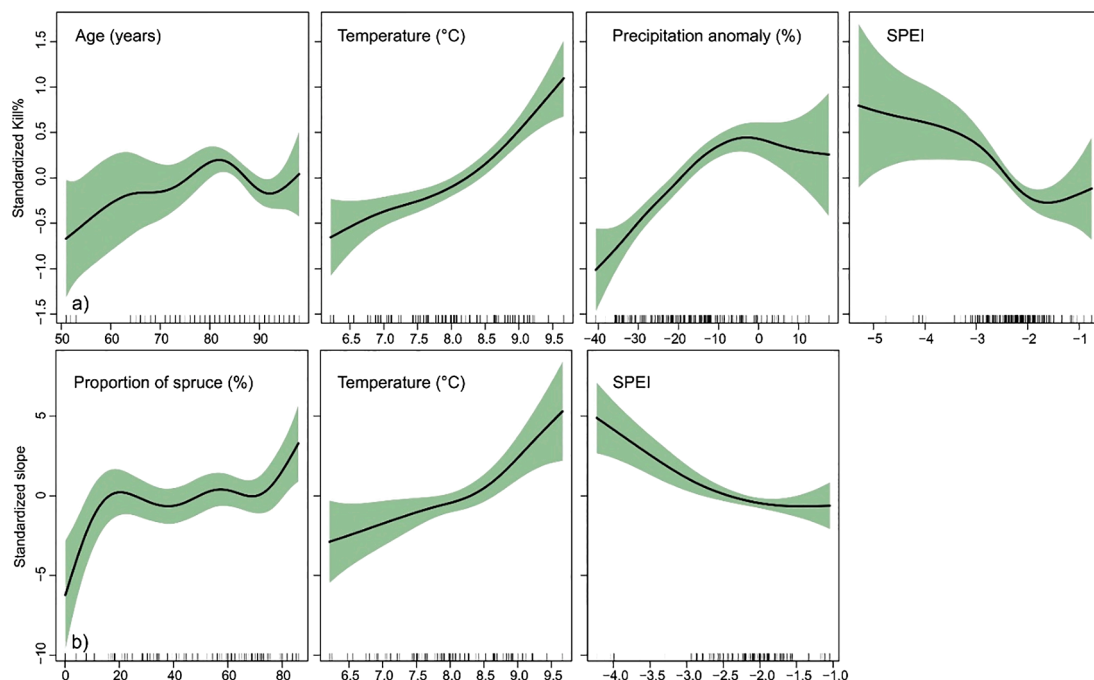


Fig. 4. Response of a) Kill% in the period 2018–2019 and b) the slope of the linear regression of Kill% to the main predictors over the period 2017–2019.

Table 1

Overview of impacts generated by the outbreak and its management, and measures taken to alleviate these effects.

Type of impact	Specific process	Quantifiable effects	Measures taken
Social and economic impacts	Reduced forest scenic and recreational values	Size of cleared areas increased from 19,000–25,000 ha (2004–2015) to 34,000 and 55,000 ha in 2018 and 2019, respectively	Reforestation applied over the area of 21,000 and 27,000 ha in 2018 and 2019, respectively; Regeneration and tending subsidized
	Restricted access to the forest	—	Salvaging applied along hiking trails to reduce the risk of injuries; Reconstruction of hiking trails
	Damage to roads by the excessive transportation of salvaged timber	At least 400 km of roads damaged	State intervention of 160 million EUR planned for 2021; Max. permissible lorry weight reduced
	Reduced quality of life of inhabitants by a high-frequency transportation of salvaged timber; risk of traffic accidents and jams	—	Communication campaigns by the Ministry of Agriculture
	Revenue loss due to the high investments into salvage harvesting, decrease in timber price, increased expenses related to the regeneration of cleared areas, and other cascading effects	Spruce log price decreased from 84 to 57 EUR between 2017 and 2019; Size of areas to be reforested increased up to 55 000 ha; Major state forest enterprise in negative balance (-30 mill. Euro in 2019); (see Appendix D for selected indicators)	Subsidy, compensation and recovery payments amounting for 259 million EUR in 2018–2019; 267 million EUR allocated for 2021
Conflicts with natureconservation	Increased use of insecticides with potential risk for the environment	The use of insecticides increased almost fourfold during the period 2017–2020, and use of alpha-cypermethrin coated nets* almost sevenfold (Forests of the Czech Republic, pers. comm.).	Recommendations on outbreak management in different protected areas (including Natura2000 sites) by the Ministry of Environment; Improved management guidelines by natural park administrations; Management recommendations by NGOs; No new legally binding regulations adopted with regard to the outbreak
	Damage to nature conservations areas from bark beetles and management activities	13,000 ha of the protected areas were affected by the outbreak (AOPK ČR, 2020)	
	Conflict about the fate of standing dead trees with no effect on bark beetle populations and low economic value	6 million and 5–10 million m ³ of dead and infested trees remained in the forest in 2018 and 2019, respectively	
Tension between post-outbreak reforestation and game management	Habitat destruction, disturbing reproduction of species of conservation concern (e.g. woodpecker <i>Dryocopus martius</i> , black grouse <i>Tetrao tetrix</i>)	—	
	Subsidized planting of seedlings on cleared areas is countered by high-intensity game browsing. High game populations are advocated by some hunting associations and other interest groups.	Approximate cost of damages from browsing 1 million EUR annually (Ministry of Agriculture of the Czech Republic, 2019); almost total devastation of non-spruce species in regeneration	Hunting season for all regeneration damaging game species extended to the whole year; Regeneration protection (repellents, fencing) subsidized; Establishment of shooting corridors in reforested areas; Motivation payments for hunters
Depletion of logistic and human resources	Management is constrained by bottlenecks in transportation and storage of salvaged wood, and production of seedlings in nurseries	1,500 railway wagons missing in 2018	Establishment of new storages for salvaged wood; Wood storage on agriculture land allowed; Enhancing railway transport capacities; Reduction of planned harvests; Nurseries adjusting their production to the new demands, millions of spruce seedlings destroyed and replaced; Relaxed roles for the geographical transfer of reproductive material; Extended reforestation period
	Excessive workload, lack of available workforce and restricted mobility of seasonal workers due to the COVID crises caused increase in labour price and lack of forestry workers	An estimated shortage of ca 6,000–10,000 forest workers annually	Simplified visa procedure for the Ukrainian workers

* A fabric net coated by an insecticide and used to cover the infested logs to prevent the beetles from leaving them (e.g. Skrzecz et al., 2015).

bird nesting (see Table 1 for quantitative indicators) (AOPK ČR, 2020). As a response, different state and non-governmental nature conservation organization developed recommendations and management guidelines on mitigating the adverse effects (e.g., Ministry of the Environment of the Czech Republic, 2019). However, no legally binding regulations beyond the scope of the existing legislation were adopted in this regard.

Another issue of critical importance concerned the devastating effect of game on forest regeneration. Investments into the planting of desired tree species on cleared areas were, in fact, cancelled out by high-intensity browsing. Systematic game reduction was, however, opposed by different interest groups. The estimated damage was found to amount for ca. 1 million EUR annually (Ministry of Agriculture of the Czech Republic, 2019). This required the adoption of unprecedented measures such as the extension of hunting season over the whole year, the establishment of shooting corridors in the reforested areas, and

payments to hunters (Table 1).

The final outbreak implication included the fast depletion of logistic and human resources. Except for depleted storage and transportation capacities, up to 10,000 forest workers were estimated to be missing annually. The measures taken included relaxed visa procedures for foreign workers, a country-wide survey of places serving as storage yards, and an increase in railway transport capacity (see Table 1).

4. Discussion

Bark beetle disturbances are increasing globally, yet some aspects of such dynamics as well as the social, market and ecological implications remain insufficiently understood (Morris et al., 2017). We investigated here the most recent outbreak of bark beetles in the Czech Republic that has exceeded ranges of disturbance impacts recorded in the managed

forests of Central Europe over the past two centuries. We found a transition from wind- to drought-driven outbreak dynamics as well as a transition from mostly locally manifested outbreaks to the regional and supranational dynamics. Our findings highlighted a low level of social preparedness to face the outbreaks (or natural disturbances in general) amplified by climate change and questioned the sustainability of production forestry oriented on Norway spruce that is broadly practiced in Central Europe. We discuss here the underlying mechanisms of the investigated outbreak dynamics, management implications, and identify the pathways of future research.

4.1. Outbreak dynamics

Most of the previous outbreaks of bark beetles in Central Europe were triggered by windthrows and occurred in geographically isolated areas (Hlásny and Turčáni, 2013; Lausch et al., 2013; Mezei et al., 2017; Modlinger and Novotný, 2015), though they were likely synchronized by climate variation and other factors acting on a large scale (Økland et al., 2005; Senf and Seidl, 2018). The ongoing outbreak deviates from this dynamic by its size and that it was not associated with any significant windthrow. Our analysis supported this observation, i.e. while the effect of wind was significant in the entire period 2003–2019, this was not the case in the recent outbreak period 2017–2019. Our findings instead suggested the prominent role of drought in unfolding large-scale epidemics. Such dynamics were described by Marini et al. (2017), who suggested that drought and climate warming can boost population eruptions even in the absence of windthrows. While these authors found no synergies between outbreak drivers, other studies found a strong amplifying effect of wind and warmer and drier climate on bark beetle dynamics (Dobor et al., 2020a, 2020b; Seidl et al., 2017). Such interactions deserve attention in future research as they may form a new disturbance dynamic in Central Europe. For example, outbreaks may be triggered by drought in hot and warm years (i.e. the incidence of a windthrow may not be a necessary precondition of the outbreak), whereas wind can trigger population transitions in cooler and moister years, cumulatively resulting in increased disturbance levels. Still, there are indications that outbreaks triggered by drought, though of a smaller extent, occurred in Central Europe also in the past (e.g., Zahradník and Zahradníková, 2019).

We found that the level of killed trees started to increase in 2014 in most of the country, and the outbreak accelerated in the remarkably dry year of 2015. Another critical acceleration corresponded with the climatically extreme year of 2018. The 2015 drought (along with the year 2003) represents the most intense drought initiated during the vegetation period over the previous 250 (Hanel et al., 2018) or 500 years (Orth et al., 2016). The 2018 drought even exceeded the severity of previous events (Schuldt et al., 2020). The 2018 increase in outbreak intensity corresponds with the fact that the highest canopy mortality over the past 34 years in Europe was observed in 2018 (Senf et al., 2020), though bark beetles represented just one factor driving this dynamic. All these events had drastic impacts on the European ecosystems, exceeding the limits for xylem hydraulic failure in many species, widespread discoloration, and premature leaf shedding (Bastos et al., 2020a, 2020b; Schuldt et al., 2020). In spruce, these events likely compromised resin exudation and the overall defense capability against bark beetles (Netherer et al., 2015), resulting in the observed outbreak accelerations. It remains unclear why bark beetle disturbance did not increase in response to the 2003 drought that particularly affected the western part of the Czech Republic (Hanel et al., 2018), while this event led to extensive forest damage across Europe (Rouault et al., 2006).

Another factor contributing to the severity and size of the outbreak was the unfavorable structure of the Czech forests that was particularly

conducive to large-scale outbreaks. Norway spruce represents 52% of species composition in the country and occurs mostly in monocultures, and vulnerable age classes prevail (Podrázský et al., 2014). Most of the spruce forests are distributed in low to medium elevations, where the risk of drought stress is high, and two to three bark beetle generations can be produced annually (Berec et al., 2013; Hlásny et al., 2011). Spruce also cannot exploit the potential buffering effect of topographic complexity in such environment (Albrich et al., 2020). At such sites, the risk of spruce damage is sevenfold higher than in its historical range (Marini et al., 2012). Moreover, spruce planted in monocultures has a significantly lower survival probability than spruce growing in species diverse stands (Griess et al., 2012; Neuner et al., 2015). Our analysis of outbreak drivers is consistent with these findings, highlighting the importance of forest age, spruce share, and air temperature as predisposing factors of infestation.

A further outbreak development remains unclear, particularly due to the importance of hardly predictable climate extremes as well as insufficient understanding of mechanisms causing outbreak collapse (Biedermann et al., 2019). Still, that the remaining spruce growing stock amounts to ca. 400 million m³ makes outbreak collapse due to the resource depletion unlikely. Instead, the current outbreak's epicenter may shift westwards after the resources are depleted, or the outbreak may expand. Moreover, climate projections consistently predict the increase in climate extremes, including drought (Grillakis, 2019). Therefore, outbreak collapse due to the persistently colder and moister climate is unlikely too. High rates of disturbance thus should be expected to persist in the coming years or decades, though weather-dependent inter-annual variation can be large.

We suggest that the coincidence of extreme climate with unfavorable forest conditions pushed the regional spruce forests to the margins of their operational space (see e.g. Keane et al., 2009; Turner et al., 2020). The current outbreak can thus be thought of as the beginning of a large-scale transformation of secondary spruce forest to other forest types, with a better match to the emergent environmental conditions. For example, Albrich et al. (2020) found in the Eastern Alps that a warming above 2 °C can trigger a shift from a conifer-dominated landscape characterized by large trees to a landscape dominated by smaller, predominantly broadleaved trees. These considerations underscore the utility of resilience framework (Johnstone et al., 2016; Scheffer et al., 2009; Turner et al., 2020) in understanding the ongoing changes in the Central European forests and their future developments.

4.2. Social and multi-actor implications

Because no systematic research on outbreak impacts existed at the time of this study, we relied on numerous public materials published by state administration, NGOs, and other bodies (e.g. Takala et al., 2019). Although such data cannot replace a sound social and economic analysis (e.g., Montagné-Huck and Brunette, 2018; Qin and Flint, 2010), they represent a sensible estimate of outbreak implications. Such an analysis can also serve as a baseline for the next assessments, helping to understand the temporal evolution of impacts and responses, which can be vital for informing future management strategies (Qin and Flint, 2010).

We found that the outbreak generated new interactions between forest management on the one side, and labor market, transportation, game management and nature conservation on the other. Most of these interactions had a negative connotation and were constraining management decisions. Such a situation partly stems from the long-term lack of participatory approaches in forestry decision-making (Hoogstra-Klein et al., 2017) and poor harmonization of sectoral policies (Fürst et al., 2017). We identified the tension between forest management and nature conservation to be particularly disturbing. Although efforts to resolve

this tension is a notorious part of the public discourse and political agenda (Maier and Winkel, 2017; Mikusiński et al., 2018), the ongoing outbreak showed that the current policies are not aligned to the emergent large-scale disturbances. Some aspects of this situation resemble the case of the Białowieża Forest Massif in Poland (Blicharska et al., 2020), where efforts to control bark beetle outbreak were not found to be aligned with local conservation objectives and EU legislation.

In this study we primarily focused on the prevalent adverse effects of the outbreak and measures taken to mitigate them. Such a picture is however, incomplete, as the outbreak has also initiated positive societal processes. For example, the loss of mature trees and related cultural services increased public awareness of the societal role of the forests and highlighted the prominence of forest adaptation to climate change. Such a tendency was evidenced by an increase in the frequency of the word *bark beetle* (in Czech) on the Internet from 7 to 30,000 annually over the period 2000–2016 to 50,000–135,000 over the period 2018–2020 (Appendix C). The outbreak also increased the involvement of the scientific community in advising the formulation of forestry policies, and promoted public involvement and volunteering (e.g. in planting the seedlings on disturbed sites). The outbreak has accelerated the transformation of forestry legislation towards increased preparedness, capacity building and better harmonization of forestry and nature conservation policies. Therefore, instant negative effects reported in this study need to be considered along with the positive effects, though their lead times can be long (e.g., the effects of increased awareness).

4.3. Management perspective

The investigated outbreak has challenged forest management by its size and rapid expansion that exceeded logistic and human resources. The bottlenecks identified throughout the entire processing chain along with the previously described multi-actor interactions define boundaries that constrain management decisions. The two major tasks that are being addressed in the context of these constraints are: (i) how to slow the outbreak down and mitigate impacts on forest owners and the entire forest-based sector, and (ii) how to regenerate the outbreak areas towards their climate adaptedness and resilience. Obviously, these short-term and forward-looking objectives compete for limited resources, and priorities differ between the stakeholders (forest owners, state administrations, timber industry vs. nature conservation, and a part of the public). The most undesired consequence of this situation would be the emergence of a new, even-aged forest cohort dominated by spruce with low resilience to future disturbances. Although a warmer climate disfavors spruce in regeneration, factors such as spruce dominance in the existing seed bank and the advanced regeneration, and the availability of spruce seedlings on the market rather than seedlings of other species (though nurseries are quickly adapting to the new demands), may compromise the resilience of the future forest (e.g., Zeppenfeld et al., 2015). Management should therefore harmonize the reforestation of cleared areas with the natural changes in the regeneration driven by climate change, reaching a higher share of climate-matching tree species in the new forest cohort (Bolte et al., 2010; Vacek et al., 2019).

The size of the outbreak investigated here highlighted the importance of a multi-scale perspective in forest management, addressing, for example, the connectivity of spruce complexes in order to prevent the large-scale spread of bark beetles (Seidl et al., 2013; Simard et al., 2012). Because a landscape-scale forest configuration is an important driver of forest resilience (Seidl et al., 2016b), neglecting this scale in outbreak management as well as the follow-up recovery actions may result in a

low resilience of future forests. However, the adoption of the landscape perspective is typically hampered by factors such as fragmented forest ownership, a poor coordination among forest owners, and a traditional focus of Central European forestry on the stand-scale (Jactel et al., 2009; Seidl et al., 2016a). The second aspect we highlight here is that the dampening effect of sanitation removal of infested trees on bark beetle dynamics was found to strongly diminish under warmer and drier climate (Dobor et al., 2020a, 2020b, 2019). This is a disturbing fact because search for and removal of infested trees aiming to control the outbreak are heavily conducted. We therefore suggest that actions aiming at the proper regeneration of post-outbreak sites and preventative measures in areas that have not yet been affected (e.g., premature harvesting, the preventative sanitation removal of infested trees, etc.) should be preferred over actions aiming at outbreak suppression in the major outbreak areas. These facts indicate that a shift of the management paradigm towards landscape-scale considerations and resilience-oriented stewardship is needed if the outbreak occurs on a large-scale and is fueled by climate change.

5. Conclusions

The Czech Republic has become an epicentre of bark beetle outbreaks that devastate Europe's forests. Our investigation showed that the current outbreak synchronously affected areas of an unprecedented size and was predominantly driven by drought, which is a new phenomenon in Central European forests. As the frequency and severity of droughts will continue to increase, such a situation is likely to occur also in the future, though the dynamics of this development remains unclear. We further showed that the outbreak has challenged all aspects of forest management and its impacts extended far beyond forestry. Negative social responses and emergent tensions between the different actors constrained forest management and required state interventions, including subsidy and recovery payments as well as changes in legislation. We suggest that although this study was conducted in the Czech Republic, it likely characterizes processes occurring at much larger territories in Central Europe and can also inform the management of disturbances other than bark beetles.

CRediT authorship contribution statement

T. Hlásny: Conceptualization, Methodology, Writing - original draft, Writing - review & editing, Formal analysis. **S. Zimová:** Resources, Data curation, Formal analysis. **K. Merganičová:** Resources, Data curation, Writing - review & editing, Formal analysis. **P. Štěpánek:** Resources, Data curation, Software. **R. Modlinger:** Resources, Data curation, Writing - review & editing. **M. Turčáni:** Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Variables investigated using the GAM-based regression modelling

See [Table A1](#)

Table A1

Response and candidate predictor variables used to investigate the drivers bark beetle dynamics in the Czech Republic.

Variable set	Variable abbreviation	Variable	Avg $\pm$ St. Dev.	Min	Max
Response variables	% Killed (2003–2019)	Percent of total growing stock killed by bark beetles (%)	1.35 ± 3.35	0.00	57.05
	% Killed (2017–2019)	Percent of total growing stock killed by bark beetles (%)	5.44 ± 6.58	0.26	57.05
Predictor variables	Slope	Slope of trend line in % Killed over the period 2017–2019	2.51 ± 2.81	−2.27	12.88
	Spruce%	Spruce proportion (%)	49.17 ± 23.59	0.15	85.98
	Age	Mean spruce age (yrs)	76.45 ± 9.94	36	98
	T	Mean annual air temperature (1980–2010) (°C)	8.01 ± 0.79	6.22	9.66
	P	Mean annual precipitation sum (1980–2010) (mm)	679.59 ± 123.82	486.93	1 100.69
	Wind	Growing stock affected by wind and salvaged (m ³)	$37,515 \pm 74,273$	27	1 091 830
	Wind ₁	Growing stock affected by wind and salvaged, one-year lag (m ³)	$36,804 \pm 75,598$	0.31	1 091 830
	Tanom	Temperature anomaly; difference between T in actual year and average T over period 1980–2010	0.69 ± 0.68	−1.08	2.13
	Panom	Precipitation anomaly; difference between P in actual year and average P over period 1980–2010	96.96 ± 15.73	58.3	154.01
	Tanom ₁	Tanom, one-year lag (°C)	0.64 ± 0.67	−1.08	2.13
	Panom ₁	Panom, one-year lag (%)	97.25 ± 16.02	58.3	154.01
	SPEI	Standardized Precipitation-Evapotranspiration Index (evapotranspiration by Thornthwaite, 1948)	$−0.46 \pm 1.17$	−5.29	2.17

Appendix B. Inter-annual variation of selected climate variables

See [Fig. B1](#)

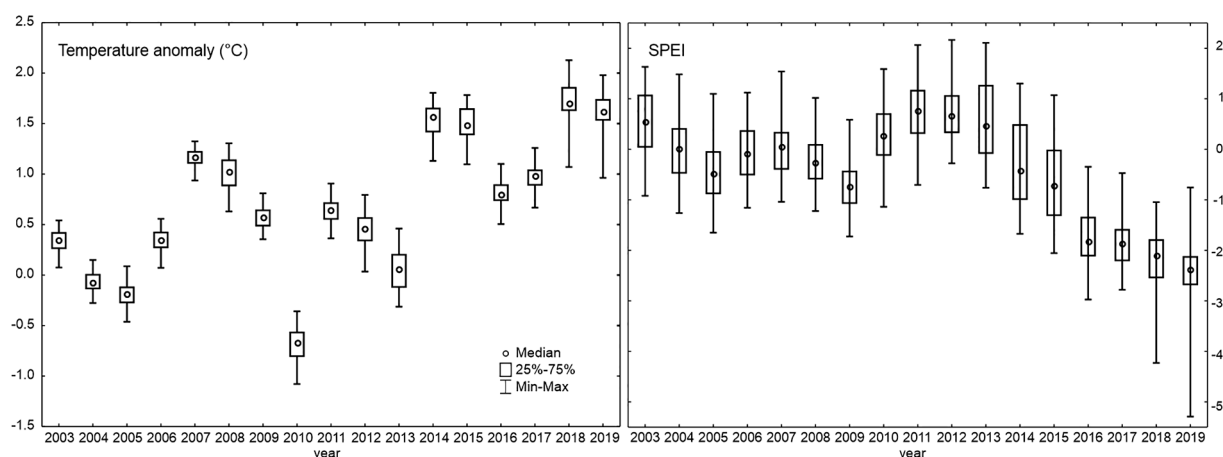


Fig. B1. Annual values of temperature anomaly (deviation from the average over the period 1980–2010) and the Standardized Precipitation-Evapotranspiration index. The range of values reflects the differences between the LAU1 districts in the Czech Republic ($n = 77$).

Appendix C. Frequency of occurrence of the Czech word “bark beetle” on the Internet

See Fig. C1

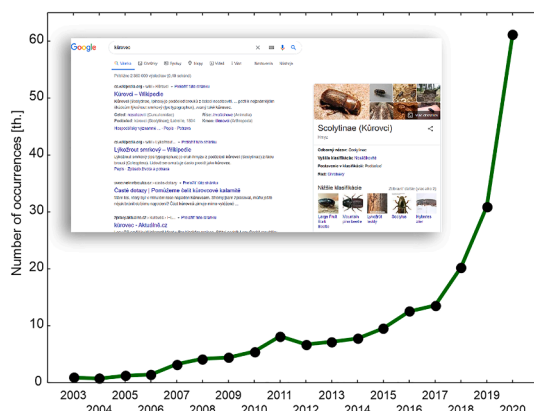


Fig. C1. The frequency of occurrence of the Czech terms for bark beetle “kůrovec” and “lýkožrout” over the period 2003–2020.

Appendix D. Selected economic indicators of the forestry sector in the Czech Republic

See Figs. D1, D2 and D3

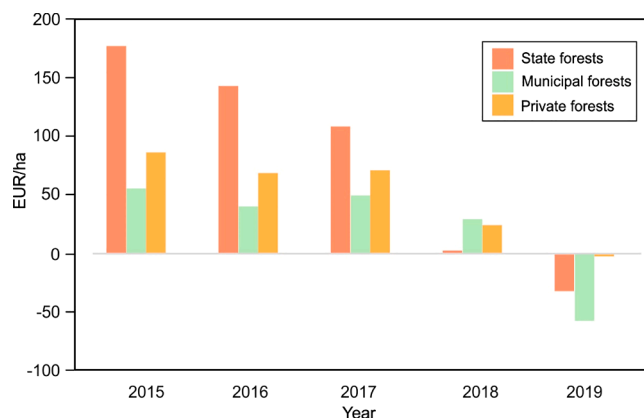


Fig. D1. Economic balance of main categories of forest owners during the recent outbreak period. The annual balance per forest area is indicated. The presented values do not include any subsidy payments (Ministry of Agriculture of the Czech Republic, 2020, 2019, 2018, 2017, 2016).

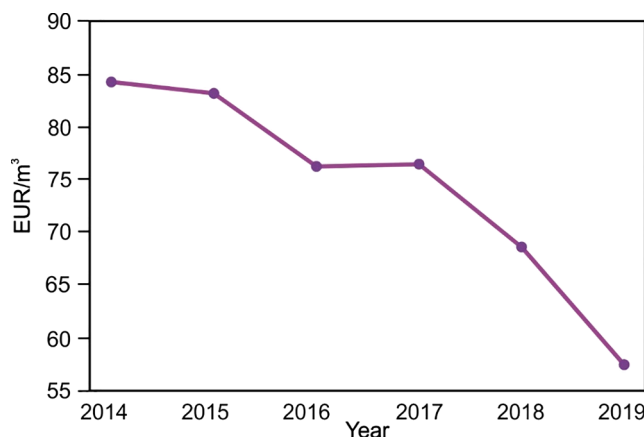


Fig. D2. Average price of spruce logs during the recent outbreak period (Ministry of Agriculture of the Czech Republic, 2020, 2019, 2018, 2017, 2016).

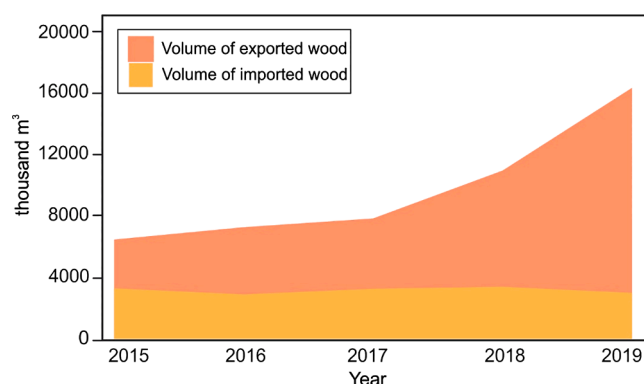


Fig. D3. Annual volume of exported and imported raw wood in the Czech Republic during the recent outbreak period (Ministry of Agriculture of the Czech Republic, 2020, 2019, 2018, 2017, 2016).

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